

Name: \_\_\_\_\_

Date: \_\_\_\_\_

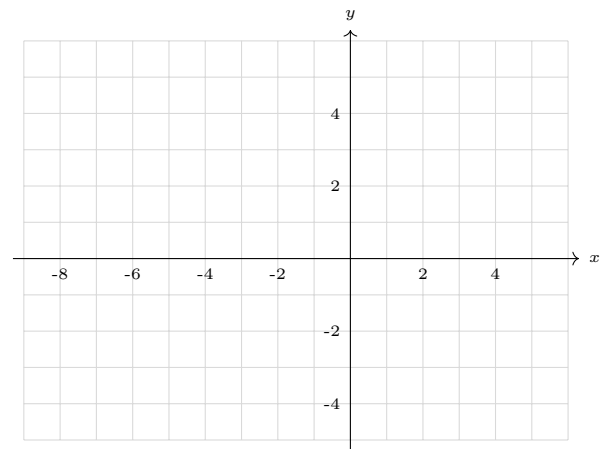
## Unit 1 Free Response #1 – Calculator Permitted

**Instructions:** Show all work. Each part is scored by the point rubric shown in the answer key.

1. Consider the piecewise-defined function  $p(x)$ :

$$p(x) = \begin{cases} -|x + 4| + 5, & -8 \leq x \leq -1 \\ (x - 2)^2 - 4, & -1 < x \leq 5 \end{cases}$$

- (2 pts) (a) Find  $p(-4)$  and  $p(0)$ . If a value does not exist, explain why.



- (4 pts) (b) Sketch an accurate graph of  $p(x)$  on the grid.

Plot at least two points for each piece and use open/closed circles correctly.

- (2 pts) (c) Explain how your graph in part (b) confirms the value of  $p(0)$  from part (a).

- (2 pts) (d) State the domain and range of  $p(x)$  in interval notation.

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## Unit 1 Free Response #2 – Calculator Permitted

**Instructions:** Show all work. Each part is scored by the point rubric shown in the answer key.

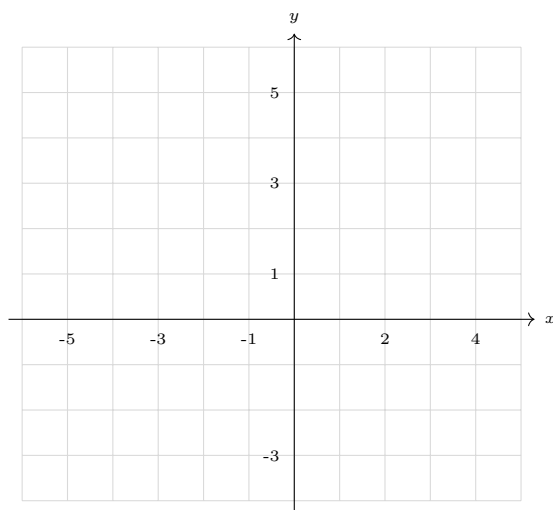
(3 pts) 1. Let  $p(x) = \sqrt{x^2 + 2x + 1}$ .

(2 pts) (a) Rewrite  $p(x)$  without a radical sign.

(1 pts) (b) Find  $p(3)$ .

(5 pts) 2. Sketch a possible graph of a function  $H(x)$  that satisfies *all* of the conditions below.

- |                                                             |              |
|-------------------------------------------------------------|--------------|
| • $H(-5)$ is undefined                                      | • $H(0) = 3$ |
| • $H(-3) = 0$                                               | • $H(2) = 5$ |
| • $H(-1) = 1$                                               | • $H(4) = 2$ |
| • Domain of $H$ : $(-5, 4]$ Range of $H$ : $(-3, 5]$        |              |
| • $H$ is increasing on $(-5, 2)$ and decreasing on $(2, 4)$ |              |

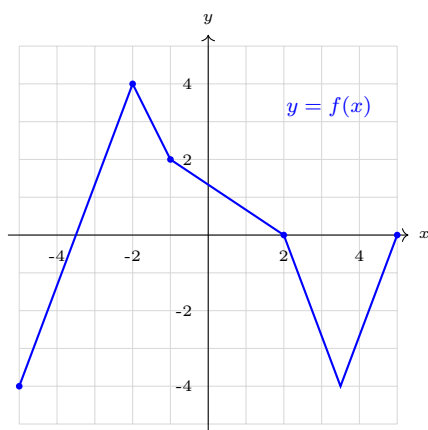


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**Unit 1 Free Response #3 – Calculator Permitted****Instructions:** Show all work. Each part is scored by the point rubric shown in the answer key.

The graph of  $f(x)$  is shown. A table of values is given for a function  $g(x)$  whose graph is line segments connecting the points.



| $x$ | $g(x)$ |
|-----|--------|
| -4  | 2      |
| -2  | -1     |
| 0   | -3     |
| 2   | -1     |
| 4   | 2      |

(2 pts) 1. Find the value of  $[-2 \cdot f(-5) - g(f(2))]$ . Show your work.

(2 pts) 2. Is  $g(x)$  even, odd, or neither? Give a numerical reason.

(3 pts) 3. If  $p(x) = ax^2 - 3x + 10$ , for what value of  $a$  is  $p(-2) = f(g(-2))$ ? Show your work.

(2 pts) 4. Identify the interval(s) of  $x$  for which  $f(x) < 0$ . Give a reason.

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## Unit 1 Free Response #4 – Calculator Permitted

**Instructions:** Show all work. Each part is scored by the point rubric shown in the answer key.

Use the tables of values for  $f(x)$  and  $g(x)$ , each with domain  $[-2, 2]$ .

| $x$    | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|--------|------|------|-----|-----|-----|
| $f(x)$ | 5    | 3    | 1   | 3   | 5   |
| $g(x)$ | $-4$ | $-2$ | $0$ | $2$ | $4$ |

(2 pts) 1. Find  $(f + g)(2)$  and  $(f \circ g)(-1)$ . Show your work.

(2 pts) 2. Determine whether  $f$  and  $g$  are even, odd, or neither. Give a numerical reason for each.

(2 pts) 3. The point  $(2, 5)$  is on the graph of  $f$ . Find the corresponding point on the graph of  $y = -f(x) + 3$ , and explain the transformation.